

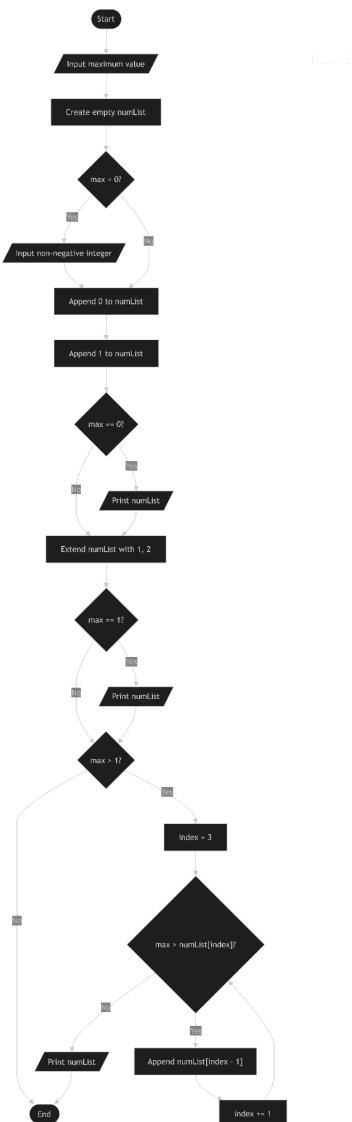
Amanda Martinez

1A)

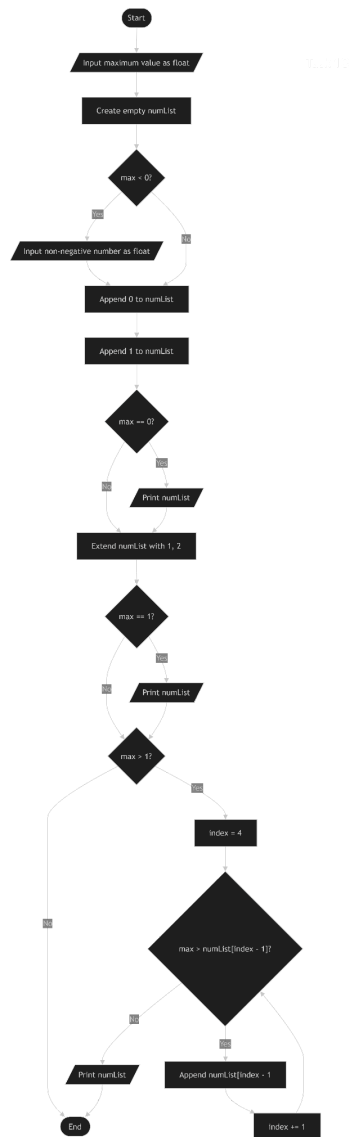
I think x will go through the values 0-5, so it'll output "Adding x to the list" 6 times for each of the values 0-5. At the end I think it will output "List consists of: 5" 5 times since that would be the last value of x.

After running the code I can say I was right about the "Adding x to the list" loop, but for the "List consists of:" loop it instead cycled through 0-5 again and output the single x value the loop was on.

1B)



1C)



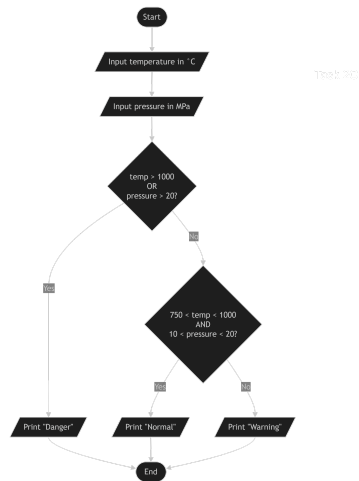
2A/B)

Logic Operations	Hand Calculation	Python Calculation
A and B	0	False
A or B	1	True
A == B	0	False
(A and B) == (A or B)	0	False
A != B	1	True
A < B	0	False

$A \leq B$	0	False
$A = B$	Error	Error

2B) There are no differences in the Python calculation, $A = B$ would be an error since a single equal sign is not comparative.

2C)

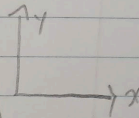


3A)

3A)

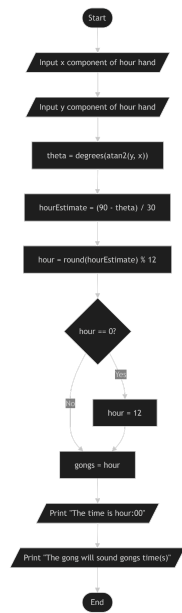
$$\vec{a}(h) = (\cos(\theta(h)), \sin(\theta(h)))$$

$$\theta(h) = 90^\circ - 30^\circ h \quad (0^\circ \text{ @ } 3:00)$$



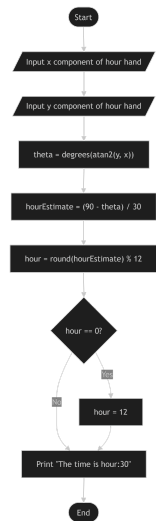
Hour	θ	Unit Vector
12:00	90°	$(0, 1)$
1:00	60°	$(\frac{1}{2}, \frac{\sqrt{3}}{2})$
2:00	30°	$(\frac{\sqrt{3}}{2}, \frac{1}{2})$
3:00	0°	$(1, 0)$
4:00	-30°	$(\frac{\sqrt{3}}{2}, -\frac{1}{2})$
5:00	-60°	$(\frac{1}{2}, -\frac{\sqrt{3}}{2})$
6:00	-90°	$(0, -1)$
7:00	-120°	$(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$
8:00	-150°	$(-\frac{\sqrt{3}}{2}, -\frac{1}{2})$
9:00	180°	$(-1, 0)$
10:00	150°	$(-\frac{\sqrt{3}}{2}, \frac{1}{2})$
11:00	120°	$(-\frac{1}{2}, \frac{\sqrt{3}}{2})$

3B)



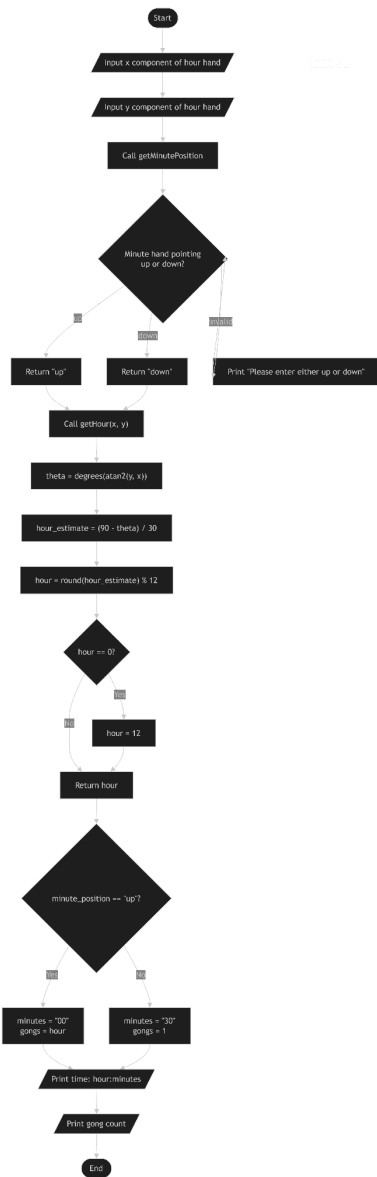
Test: 3C

3C)

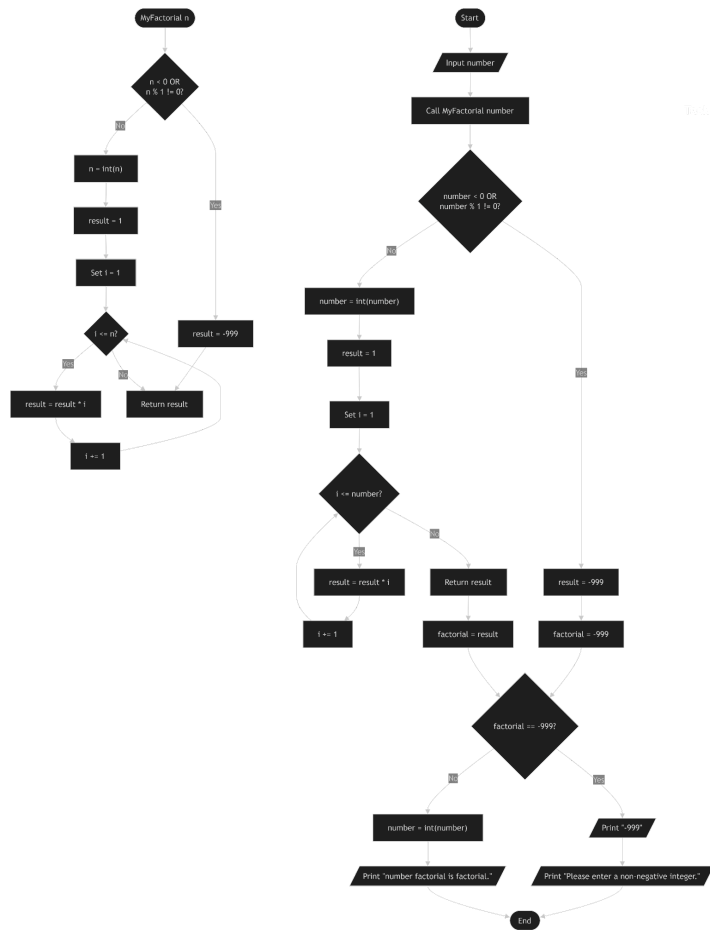


Test: 3D

3D)



4A)



5)

